HW4

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CSP 1

The eight edges are

$E = \{ab, ac, ad, bc, be, cd, ce, de\}$, $x_u \neq x_v \ (uv \in E)$

After applying the unary constraints:

$D(a) = \{1,2\}$, $D(b) = \{1,2,3\}$, $D(c) = D(d) = D(e) = \{1,2,3,4\}$

Selection rules. Interpret minimum value heuristic as minimum remaining values (MRV): choose the unassigned variable with the smallest current domain. Break MRV ties by the largest number of unassigned neighbors (degree heuristic), then alphabetically. Try values in increasing order. After each assignment, use forward checking to remove its value from every unassigned neighbor's domain.

step	Assignment	D(a)	D(b)	D(c)	D(d)	D(e)
0	—	{1,2}	{1,2,3}	{1,2,3,4}	{1,2,3,4}	{1,2,3,4}
1	a=1		{2,3}	{2,3,4}	{2,3,4}	{1,2,3,4}
2	b=2			{3,4}	{2,3,4}	{1,3,4}
3	c=3				{2,4}	{1,4}
4	d=2					{1,4}
5	e=1					

A dash in a domain column means that the variable is already assigned.

1. set a=1, delete 1 from D(b), D(c), D(d).

2. Set b=2; delete 2 from D(c); D(e)

3. Set c=3; delete 3 from D(d), D(e)

4. choose d, alphabetically. Set d=2. Since $2 \notin D(e)$, D(e) does not change.

5. Set e=1.

$$(a, b, c, d, e) = (1, 2, 3, 2, 1)$$

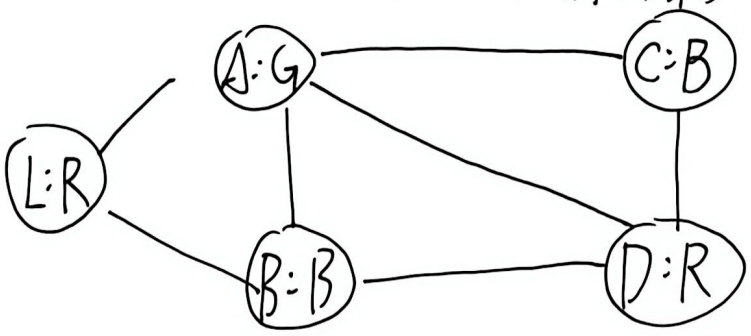
For edges, values are

$((1,2), (1,3), (1,2), (2,3), (2,1), (3,2), (3,1), (2,1))$.

CSP2 Label the left node L upper-left A, lower-left B, upper-right C, and lower-right D. The edges are LA, LB, AB, AC, AD, BD, CD

since L=R the adjacent nodes A, B cannot be red. Also $A \neq B$, so choose A = G, B = B.K

$$(L, A, B, C, D) = (R, G, B, B, R)$$



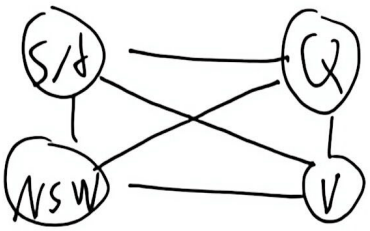
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CSP3

set. SA: South Australia, Q = Queensland, NSW: New South Wales.

V: Victoria, WA: Western Australia, NT = Northern Territory.

{SA-Q, SA-NSW, SA-V, Q-NSW, Q-V, NSW-V}.



These four regions form K_4. With only three colors at least two would have the same color, contradicting their adjacency. Therefore

Four colors also suffice. One valid coloring is

Region	WA	NT	SA	Q	NSW	V
Color	2	3	1	2	3	4

Hence $\chi \leq 4$, and

$$\chi = 4$$

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